Branch/Class/Batch: AI&DS/TE/A1
Student Roll no.: A16
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EXPERIMENT NUMBER: 4

AIM: Simulate sliding window protocols using any packet tracer tool (any one from Stop and Wait, Go-Back-N, Selective Repeat)

OBJECTIVES: To simulate any one Sliding Window Protocol using a network simulation tool and study its data transmission and flow control mechanism.

THEORY:

1. Differentiate between Stop-and-Wait, Go-Back-N, and Selective Repeat Protocols.
2. What is flow control in computer networks?

Sliding Window Protocol

The **Sliding Window Protocol** is a **flow control protocol** used in computer networks to ensure efficient and reliable transmission of data between a sender and a receiver. Instead of waiting for an acknowledgment after sending each frame, the sender is allowed to transmit multiple frames before receiving acknowledgments. The number of frames that can be transmitted at a time is determined by the **window size**. As acknowledgments are received, the transmission window moves (slides) forward, allowing new frames to be sent. This technique improves network throughput and reduces transmission delays.

Flow Control

Flow Control is a technique used to regulate the rate at which data is transmitted from the sender to the receiver. It prevents the sender from overwhelming the receiver with excessive data, ensuring that the receiver has enough time and buffer space to process incoming packets. Flow control improves communication reliability, prevents data loss, and enhances network performance.

Types of Sliding Window Protocols

1. Stop-and-Wait Protocol

Stop-and-Wait is the simplest flow control protocol. The sender transmits one frame and waits for an acknowledgment (ACK) before sending the next frame. If the acknowledgment is not received within a specified time, the sender retransmits the frame. Although simple and reliable, it has low efficiency because the sender remains idle while waiting for acknowledgments.

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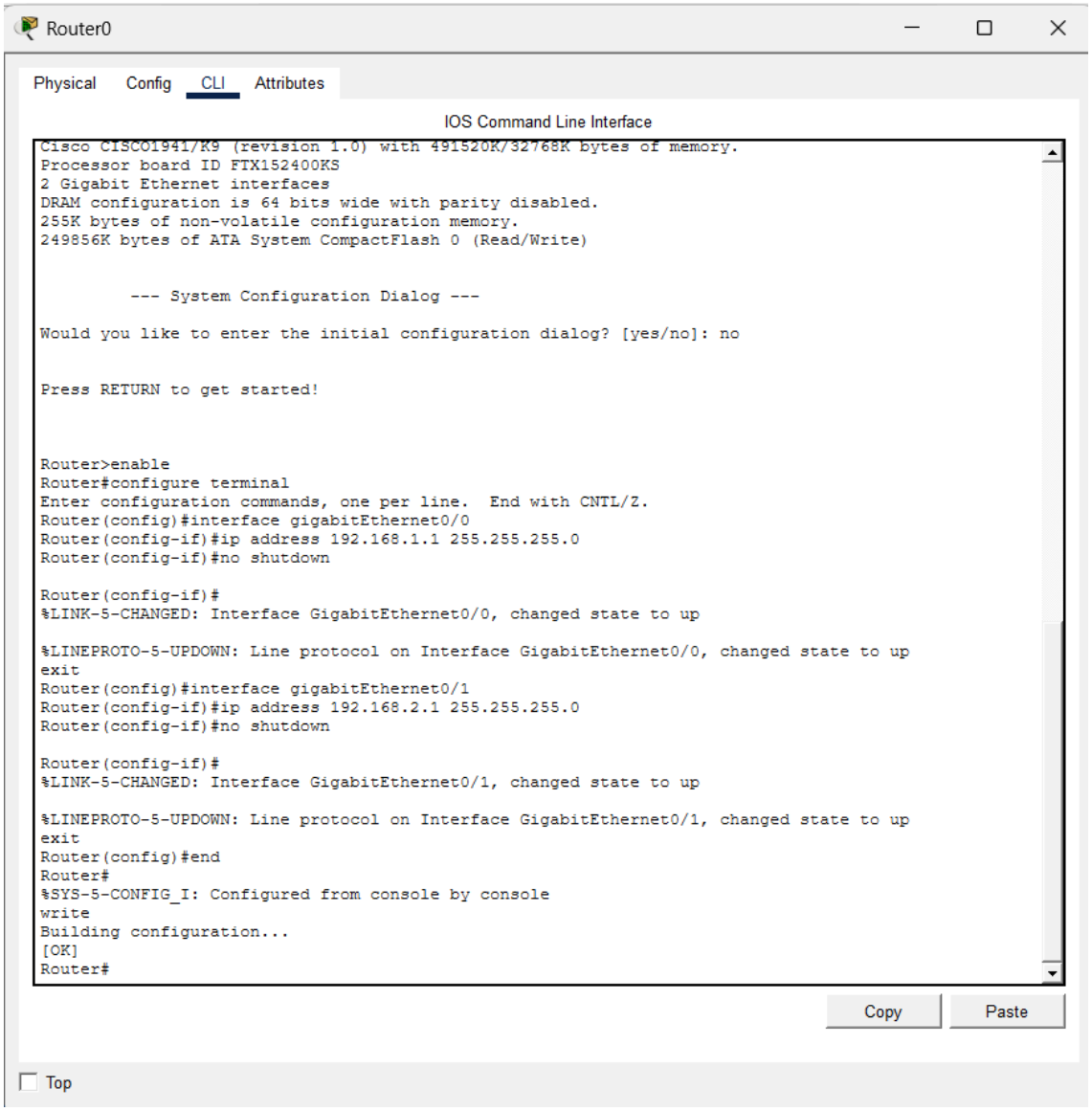
2. Go-Back-N Protocol

Go-Back-N allows the sender to transmit multiple frames continuously without waiting for individual acknowledgments. If an error occurs or a frame is lost, the receiver discards that
frame and all subsequent frames. The sender then retransmits the lost frame along with all the following frames. This protocol provides better efficiency than Stop-and-Wait but may require retransmission of correctly received frames.

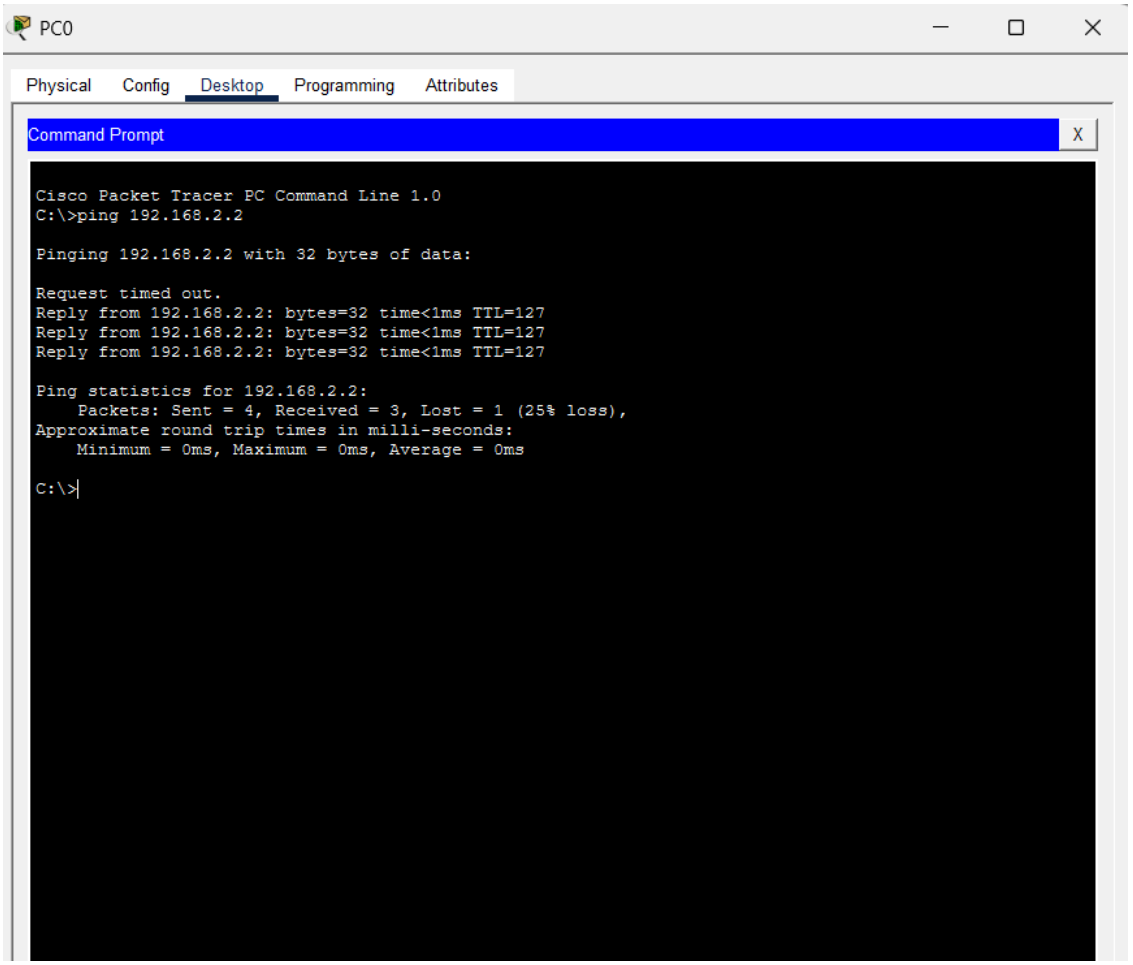
3. Selective Repeat Protocol

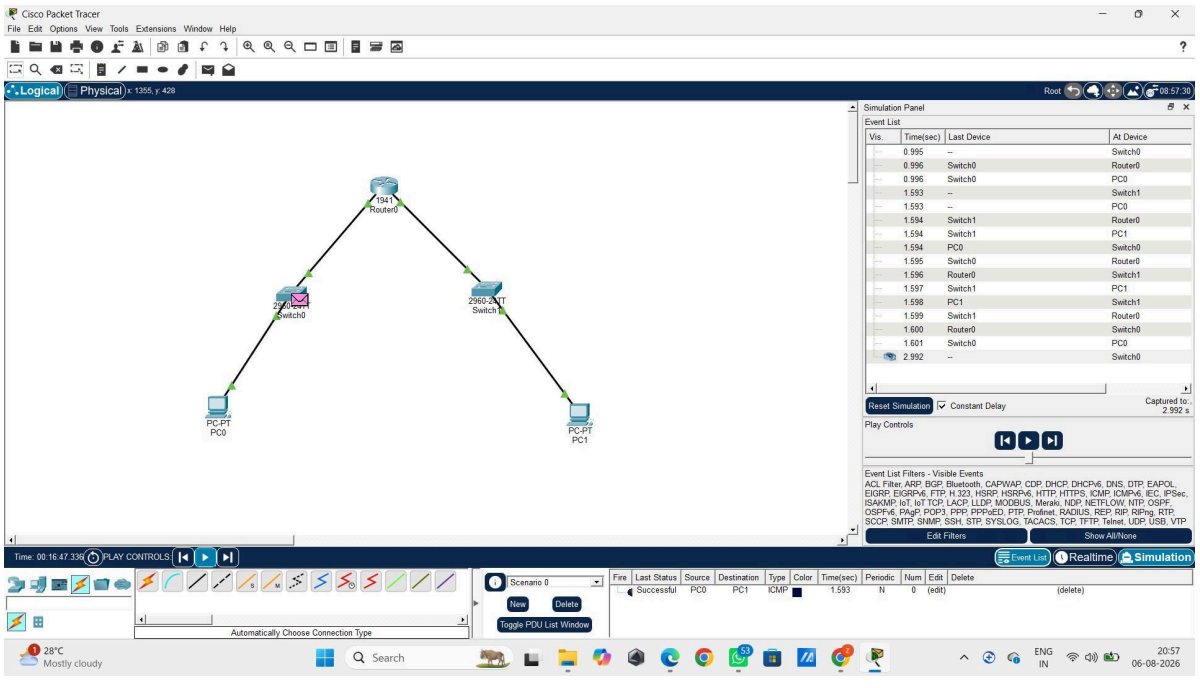
Selective Repeat is an advanced sliding window protocol in which only the lost or corrupted frames are retransmitted instead of retransmitting all subsequent frames. The receiver stores correctly received frames even if some frames are missing. This protocol offers higher efficiency and better bandwidth utilization but requires more memory and processing.

DIAGRAM:



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RESOURCES REQUIRED:

- Cisco Packet Tracer Software
- Computer/Laptop
- 2 PCs (PC0 and PC1)
- 2 Cisco 2960 Switches
- 1 Cisco Router (1941/2911)
- Copper Straight-Through Cables

PROCEDURES:

Procedure

1. Open **Cisco Packet Tracer**.
2. Place the following devices in the workspace:
    - 2 PCs (PC0 and PC1)
    - 2 Switches (Switch0 and Switch1)
    - 1 Router (Router0)
3. Connect the devices using **Copper Straight-Through** cables as shown below:

    PC0 —— Switch0 —— Router0 —— Switch1 —— PC1

4. Assign IP addresses to the PCs and router interfaces.

PC0

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

PC1

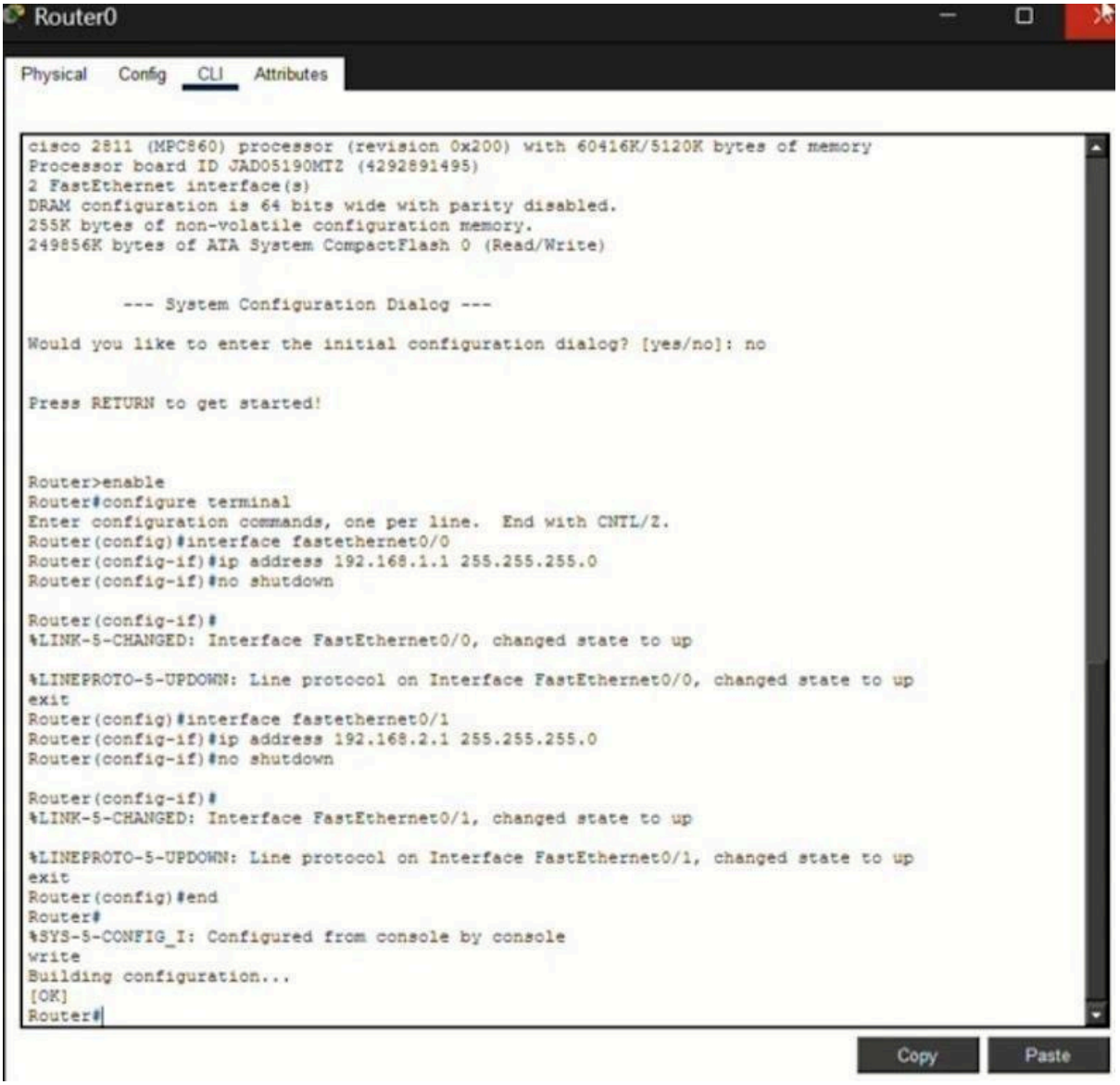
- IP Address: 192.168.2.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

Router0

- **GigabitEthernet0/0**
    - IP Address: 192.168.1.1
    - Subnet Mask: 255.255.255.0
- **GigabitEthernet0/1**
    - IP Address: 192.168.2.1
    - Subnet Mask: 255.255.255.0

5. Configure the router interfaces and enable them using the following commands:

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6. Verify the physical connectivity by checking that all links turn **green**.
7. Test network connectivity by opening the **Command Prompt** on **PC0** and executing:

   ```
   ping 192.168.2.2
   ```

   Verify that replies are received successfully.

8. Switch from **Real-Time Mode** to **Simulation Mode**.
9. In the **Event List Filters**, select **ARP** and **ICMP** packets.
10. Generate traffic by sending a **Simple PDU** or by executing the **Ping** command from PC0 to PC1.
11. Click **Capture/Forward** repeatedly and observe the movement of each packet through:
    - PC0
    - Switch0
    - Router0

- Switch1
- PC1

RESULT:

The sliding window protocol simulation environment was successfully created using Cisco Packet Tracer. Communication between two different networks was established through a router. The **PING** utility verified successful connectivity, and **Simulation Mode** was used to observe ARP and ICMP packet transmission, demonstrating the basic concepts of flow control and packet forwarding.

CONCULSION:

The experiment was successfully performed using Cisco Packet Tracer. A routed network was configured with two PCs, two switches, and one router. IP addresses and router interfaces were configured correctly, and communication between the two networks was verified using the **PING** utility. Packet transmission was observed in **Simulation Mode**, providing a practical understanding of flow control, packet forwarding, and the operation of Sliding Window Protocol concepts in a simulated network environment.